

Midterm Exam 1 · Fall 2022

MATH 411 · October 7, 2022

Retypeset past exam for practice. The original allowed one handwritten page of notes, lasted 50 minutes, and was worth 25 points. The original date and Question 3 contained errors corrected here.

1. (a) **(3 points)** Find every solution of $z^3 = -8i$. You may leave your answers in polar form $re^{i\theta}$.
(b) **(4 points)** Let $r > 0$ be real and $w \in \mathbb{C}$. Show that

$$\left| \frac{w - r}{1 - r\bar{w}} \right| \leq 1 \quad \text{when } r < 1 \text{ and } |w| < 1.$$

2. **(9 points)** Let $D = \{z \in \mathbb{C} : |z| < 1\}$ and let $f = u + iv$ be holomorphic on D . Suppose $|f|^2 = u^2 + v^2 = c$ throughout D , where $c > 0$. Prove that f is constant.

Hint. Differentiate $u^2 + v^2 = c$ with respect to x and y . You may use that a real function with continuous partial derivatives and both partial derivatives zero on D is constant.

3. **(9 points)** Let f be entire and analytic, and set

$$A_n = \{z \in \mathbb{C} : f^{(n)}(z) = 0\}, \quad n \geq 0.$$

- (a) Show that A_n is closed. Show also that A_n is discrete if $f^{(n)}$ is not identically zero. What is A_n if $f^{(n)}$ is identically zero?

Here *discrete* means that every point of the set has a neighborhood containing no other point of the set.

- (b) Suppose that for every $a \in \mathbb{C}$, at least one coefficient in the Taylor expansion

$$f(z) = \sum_{n \geq 0} c_n(a)(z - a)^n$$

is zero. Show that f is a polynomial. You may use part (a), the fact that closed discrete subsets of $\mathbb{C}$ are countable, and the fact that $\mathbb{C}$ is uncountable.